BIO 005 Human Physiology · Week 8 notes · Dr. Sharilyn Rennie

Reproductive hormones

The written notes for the reproductive hormones walkthrough. Week 8 competencies 16 to 21.

To print: these notes as a two-column PDF. Already printed the Week 8 notes? You do not need to print these again.

How to use these notes. Work through the walkthrough first, writing each prediction on your worksheet before you press Show me. Then read these notes to fill in anything you missed, and use them to review. They are the same facts in a form you can study from, not a copy of the walkthrough.

Step by step: the hormone sequences

In your textbook: Silverthorn Chapter 26, the parts on male reproduction, female reproduction and the menstrual cycle, and pregnancy, labor and lactation.

The male axis, step by step

1. Neurons in the hypothalamus release GnRH in pulses into the portal vessels.
2. GnRH reaches the anterior pituitary, where the gonadotrophs release LH and FSH into the blood.
3. LH reaches the Leydig cells, between the seminiferous tubules, and they make testosterone.
4. Testosterone diffuses into the nearby tubules (a paracrine signal) and also enters the blood.
5. FSH reaches the Sertoli cells inside the tubules. With testosterone, they support spermatogenesis and make androgen-binding protein, which keeps testosterone high around the developing sperm.
6. The Sertoli cells also make inhibin.
7. Testosterone feeds back on the hypothalamus (less GnRH) and the anterior pituitary (less LH). Inhibin feeds back on the anterior pituitary and reduces FSH.

The ovarian cycle, step by step (a 28 day cycle)

1. Day 1 is the first day of menstrual bleeding. FSH rises a little, and a small group of follicles starts to grow.
2. LH drives the theca cells to make androgens. FSH drives the granulosa cells to convert them to estradiol with aromatase.
3. Moderate estradiol, with inhibin, is negative feedback: GnRH, LH and FSH stay low.
4. As FSH falls, one dominant follicle keeps growing and the others break down.
5. The dominant follicle makes more and more estradiol. When estradiol stays high for about two days, its feedback flips to positive.
6. GnRH rises and the pituitary releases the LH surge, with a smaller rise in FSH.
7. The LH surge makes the egg finish its first meiotic division and the follicle rupture. Ovulation happens roughly a day and a half after the surge begins, around day 14.
8. The ruptured follicle becomes the corpus luteum, which makes progesterone, estrogen and inhibin. Feedback is negative again, and LH and FSH stay low.
9. Without a pregnancy, the corpus luteum breaks down after about 12 days. Progesterone and estrogen fall, and the next cycle begins.

The uterine cycle, step by step

1. Menstrual phase: the functional layer is shed with blood, and the basal layer stays.
2. Proliferative phase: estradiol from the growing follicles makes the functional layer thicken, the glands lengthen and the spiral arteries grow up into it. Estrogen also adds progesterone receptors.
3. Secretory phase: after ovulation, progesterone from the corpus luteum makes the glands coil and secrete, and the spiral arteries coil more.
4. When the corpus luteum dies, progesterone falls, the spiral arteries constrict, and the functional layer is starved of oxygen and dies.
5. The arteries relax, blood flows into the dying tissue, and the functional layer is shed. This is day 1 of the next cycle.

Pregnancy, step by step

1. The embryo implants in the lining about a week after ovulation.
2. Its outer cells, the trophoblast, release hCG.
3. hCG binds the LH receptors on the corpus luteum and keeps it alive and making progesterone and estrogen.
4. Progesterone stays high, so the lining is kept and there is no period.
5. Around weeks 7 to 9, the placenta takes over most of the progesterone production, and the corpus luteum is no longer needed.

Labor, step by step (positive feedback)

1. The fetal head presses on the cervix and stretches it.
2. Stretch receptors in the cervix send signals to the hypothalamus.
3. Oxytocin neurons release oxytocin from the posterior pituitary.
4. Oxytocin makes the uterine muscle contract, helped by prostaglandins made in the uterus.
5. Stronger contractions push the head harder into the cervix, so the stretch increases and the loop repeats, stronger each time.
6. Delivery removes the stretch, and the loop stops.

Suckling, step by step

1. The baby suckles, and nerve signals travel from the nipple to the hypothalamus.
2. Prolactin from the anterior pituitary rises (suckling reduces the dopamine that normally holds it down). Prolactin makes the alveolar cells produce milk, mostly for the next feeding.

3. Oxytocin from the posterior pituitary makes the myoepithelial cells around each alveolus contract, and milk is pushed into the ducts: the let-down.

The gonadal axis

In your textbook: Silverthorn Chapter 26, the part on basic patterns of reproduction, on hormonal control.

Dr. Rennie's videos for this part: The gonadotropins: FSH and LH.

Reproduction follows the same hypothalamic-pituitary pattern as the other axes: a releasing hormone from the hypothalamus, a trophic hormone from the anterior pituitary, and a hormone from the target gland that feeds back.

GnRH

Gonadotropin-releasing hormone, from the hypothalamus, carried in the portal vessels to the anterior pituitary.

Gonadotropins

LH (luteinizing hormone) and FSH (follicle-stimulating hormone), released by the gonadotrophs of the anterior pituitary. They act on the gonads, the testes and the ovaries.

Pulsatile GnRH

GnRH is released in bursts. The pituitary needs the pulses: steady GnRH down-regulates its GnRH receptors, and after a brief rise LH and FSH fall. Drugs that act like GnRH, given continuously, are used to shut down the gonads, for example to lower testosterone in some prostate cancers.

Gonad	LH acts on	Which makes	FSH acts on	Which does
Testis	Leydig cells, between the tubules	Testosterone	Sertoli cells, inside the tubules	Supports spermatogenesis; makes androgen-binding protein and inhibin
Ovary	Theca cells, the outer layer of the follicle	Androgens	Granulosa cells, the inner layer	Converts androgens to estradiol (aromatase); makes inhibin

- In both sexes, LH drives the cell that makes the steroid and FSH drives the cell that supports the gamete.
- In both sexes, the FSH-driven cell makes inhibin, which feeds back on FSH alone.

The testis

In your textbook: Silverthorn Chapter 26, the part on male reproduction, on spermatogenesis and its hormonal control.

- Sperm are made in the seminiferous tubules. Spermatogonia sit at the outer wall; spermatocytes and spermatids mature as they move inward; sperm are released into the lumen. This takes a little over two months.
- Sertoli cells run from the wall to the lumen, wrap the germ cells and support them.
- Leydig cells sit outside the tubules, beside blood vessels, and make testosterone.
- Testosterone, a steroid, diffuses into the tubules as a paracrine signal and also enters the blood.
- Androgen-binding protein, made by Sertoli cells under FSH, binds testosterone and keeps it concentrated around the developing sperm.
- Spermatogenesis needs both FSH and testosterone acting on the Sertoli cells.

Blood–testis barrier

Tight junctions between neighboring Sertoli cells, near the outer wall. They separate an outer compartment, open to the blood, holding the spermatogonia, from an inner compartment where later germ cells develop. Developing sperm carry proteins that first appear at puberty, so the barrier keeps immune cells and antibodies from attacking them. It also lets Sertoli cells control the fluid around the sperm.

Hormone	Made by	Feeds back on	Effect
Testosterone	Leydig cells	Hypothalamus and anterior pituitary	Less GnRH and less LH
Inhibin	Sertoli cells	Anterior pituitary	Less FSH

Testosterone from outside

1. High blood testosterone from a shot or anabolic steroids acts as negative feedback.
2. GnRH, LH and FSH fall.
3. Without LH, the Leydig cells stop making testosterone, so the very high level inside the testis falls, even though the blood level is high.
4. Without FSH and local testosterone, Sertoli cells cannot support spermatogenesis. Sperm production falls and the testes can shrink.

The ovarian cycle

In your textbook: Silverthorn Chapter 26, the part on female reproduction and the menstrual cycle, on the ovarian cycle and its hormonal control.

Phase	Days (28 day cycle)	In the ovary	Hormones
Follicular	1 to 14	A group of follicles grows; one becomes dominant	FSH up a little early; estradiol rises; moderate estradiol and inhibin are negative feedback
Ovulation	About 14	The LH surge ruptures the mature follicle and releases the egg	High, sustained estradiol is positive feedback; LH surges, FSH rises a little
Luteal	14 to 28	The corpus luteum forms and, without pregnancy, breaks down after about 12 days into the corpus albicans	Progesterone peaks about a week after ovulation; estradiol has a second, smaller peak; LH and FSH held low

Follicle

An oocyte surrounded by granulosa cells, with a ring of theca cells outside.

Dominant follicle

The one follicle that keeps growing as FSH falls, the one with the most FSH receptors and the best blood supply.

LH surge

A large burst of LH set off when estradiol stays high (for about two days). Positive feedback: high, sustained estradiol makes the hypothalamus and anterior pituitary release more GnRH and LH instead of less. Progesterone, beginning to rise from the follicle, also helps.

Ovulation

Rupture of the follicle and release of the egg, roughly a day and a half (about 32 to 36 hours) after the LH surge begins. The surge also makes the egg finish its first meiotic division. The egg can be fertilized for only about a day.

Corpus luteum

The yellowish gland formed from the ruptured follicle. It makes progesterone, estrogen and inhibin, which hold LH and FSH low so no new follicle matures.

- The surge ends on its own: LH starts turning the follicle cells toward making progesterone, so estradiol falls even before the follicle ruptures, and the pituitary becomes less responsive. After ovulation the follicle is gone.
- The luteal phase is fairly constant at about 14 days because the corpus luteum's lifespan is fixed. The follicular phase is the part that varies. A 32 day cycle ovulates around day 18.
- Counting back 14 days from the next period predicts ovulation better than counting forward from the last one.

Reading the cycle graph

- The LH surge places ovulation: ovulation comes after it.
- The estradiol peak comes about a day before the LH surge and sets it off. It comes from the dominant follicle.
- The progesterone peak, about a week after ovulation, comes from the corpus luteum.
- A progesterone rise in the second half of the cycle proves ovulation happened. In a cycle without ovulation, the LH surge is missing and progesterone never rises.

The uterine cycle

In your textbook: Silverthorn Chapter 26, the part on female reproduction and the menstrual cycle, on the uterine cycle.

Endometrium

The lining of the uterus. The basal layer stays; the functional layer grows each cycle and is shed if no embryo implants. Glands run through it, and spiral arteries supply it.

Myometrium

The smooth muscle of the uterus, under the endometrium.

Uterine phase	Ovarian phase it matches	Driving hormone and source	What the lining does
Menstrual	Start of the follicular phase	Loss of progesterone (corpus luteum dies)	Spiral arteries constrict, the functional layer dies and is shed
Proliferative	Follicular	Estradiol from the growing follicles	Thickens; glands lengthen; spiral arteries grow; progesterone receptors added
Secretory	Luteal	Progesterone (with estrogen) from the corpus luteum	Glands coil and secrete glycogen-rich fluid; arteries coil more; ready for an embryo

- Around ovulation, estrogen makes the cervical mucus thin and watery, which helps sperm through. Progesterone makes it thick and sticky, a plug against bacteria and sperm.
- The lining is lost because progesterone goes away, not because a hormone arrives. If the corpus luteum is rescued and progesterone stays up, the lining stays.

Pregnancy

In your textbook: Silverthorn Chapter 26, the part on pregnancy, on implantation and the hormones of pregnancy.

hCG

Human chorionic gonadotropin, released by the trophoblast (the outer cells of the embryo) after implantation, about a week after ovulation. It is similar in structure to LH and binds the LH receptors already on the corpus luteum, keeping it alive and making progesterone and estrogen. It passes into the urine, where pregnancy tests detect it.

Rescue of the corpus luteum

Because hCG keeps the corpus luteum going, progesterone stays high, the lining is kept and there is no period.

Luteal–placental shift

Around weeks 7 to 9 (sources differ slightly; by the end of the first trimester the placenta is the main source), the placenta takes over most progesterone production. If this failed and the corpus luteum died on schedule, progesterone would fall, the lining would shed, and the pregnancy would be lost.

Hormone	Main source in later pregnancy	Role
hCG	Trophoblast and placenta	Rescues the corpus luteum; peaks in the first trimester, then falls
Progesterone	Placenta	Keeps the uterine muscle quiet and maintains the lining; keeps the breast from making milk
Estrogen	Placenta	Grows the uterus and the breast ducts; late in pregnancy adds oxytocin receptors to the uterine muscle
Relaxin	Corpus luteum and placenta	Thought to soften the cervix and loosen the pelvic ligaments for birth; much of what is known comes from animal studies

Labor and milk

In your textbook: Silverthorn Chapter 26, the part on labor and delivery, and the part on lactation.

- Late in pregnancy, estrogen increases oxytocin receptors on the uterine muscle, and the cervix softens.
- Labor is positive feedback: cervical stretch, then oxytocin, then stronger contractions, then more stretch. Prostaglandins from the uterus add to the contractions.
- A positive feedback loop runs until its stimulus is removed. Delivery removes the stretch and ends the loop.
- Synthetic oxytocin is used to start labor or strengthen contractions.

	Milk production	**Milk ejection (let-down)**
Hormone	Prolactin	Oxytocin
Released from	Anterior pituitary	Posterior pituitary
Acts on	Alveolar cells	Myoepithelial cells around the alveoli
Effect	Make milk, mostly for the next feeding	Contract and push milk into the ducts
Control	Mainly inhibited by dopamine from the hypothalamus; suckling reduces the inhibition	Released by suckling; can become a learned response, such as to a baby's cry; stress can inhibit it

- During pregnancy, estrogen, progesterone and prolactin prepare the breast, but high progesterone keeps it from making milk. After delivery, progesterone falls and milk production begins.
- A mother can make plenty of milk and still not deliver it: production (prolactin) is working, but ejection (oxytocin) is held back, as stress can do.